Library Hallucinations in LLM-Generated Code: A Risk Analysis Grounded in Developer Queries

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Motivation

→ LLMs continue to hallucinate, even when generating code.

→ Code hallucinations ***create reliability and security risks***.

→ Whilst well studied, there is no understanding of how realistic developer queries can impact the hallucination rates.

→ We test how prompt variations, such as user-inspired descriptions and mistakes, trigger library hallucinations.

Methodology

→ ***7 diverse LLMs*** evaluated, inc. GPT, Claude, DeepSeek.

→ LLMs solve Python ***BigCodeBench*** tasks with...

↳ ***Developer-style descriptions*** from StackExchange.

↳ Generated ***library misspellings and fabrications***.

→ Then imports and members are extracted from generated code and validated against PyPI and documentation.

When LLMs generate library-oriented code...

They often ignore descriptive adjectives.

Requests for “fast”, “simple”, “modern”, or “lightweight” libraries rarely changed library choices or triggered hallucinations.

They hallucinate under recency pressure.

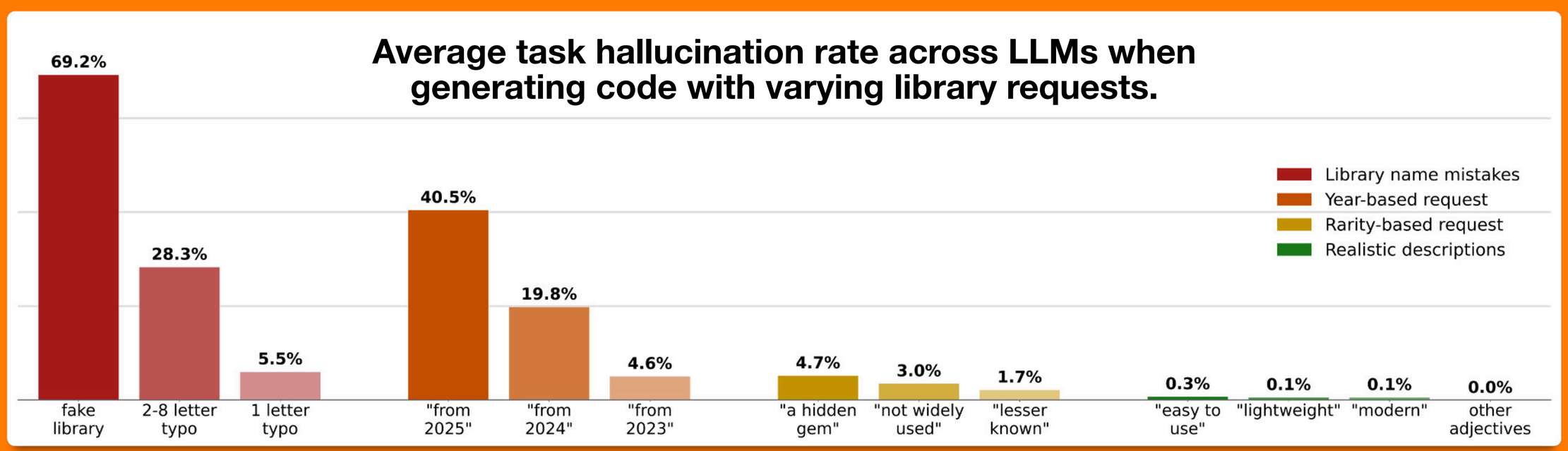
Year-based prompts caused sharp increases in hallucinated libraries, reaching up to 85% of tasks when asking for a library “from 2025”.

They comply with small library-name errors.

Even one-character typos were sometimes treated as valid libraries, causing hallucinations in up to 26% of tasks.

They consistently use fabricated libraries.

Fabricated library names were used in up to 99% of tasks, showing strong compliance with invalid user references.



Additional Results

→ Rarity-seeking prompts increase hallucinations.

→ Similar risks appear across ***JavaScript*** and ***Rust***.

→ Prompt engineering helps inconsistently.

→ Tool access can reduce hallucinations, but models can still misuse or underuse it.

Discussion

→ Library hallucinations create practical supply chain risks through ***slopsquatting*** and ***typosquatting***.

→ Temporal requests require ***cut-off-aware behaviour***.

→ Incorrect libraries require ***anti-sycophancy behaviour***.

→ We release ***LibHalluBench*** to test future models.

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